UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2020/2021

MAY 2020 TRIMESTER

FINAL ASSESSMENT

UECS3253/ UECS3463 WIRELESS APPLICATION DEVELOPMENT

SATURDAY, 3 OCTOBER 2020 TIME : 1.30 PM – 5.00 PM (3.5 HOURS)

BACHELOR OF SCIENCE (HONOURS) SOFTWARE ENGINEERING
BACHELOR OF ENGINEERING (HONOURS) ELECTRONICS
(COMPUTER NETWORKING)

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an individual, time restricted assessment which consists of THREE (3) questions. For Q1 & Q2 consist of 33 marks and Q3 consist of 34 marks.

2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **5.00 PM, 3 OCTOBER 2020.**

3. During the period of [**3.5 hours**] of this FA, the examiner(s) can be reached at
   (a).WBLE or Microsoft Teams with Code/Password: cql8u1g, or
   (b). Email: farizuwana@utar.edu.my
   You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.

4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Proper referencing is necessary to avoid plagiarism. The preferred referencing style is American Psychological Association (APA) / Harvard / Chicago Manual. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

Answer Script File

5. The answer script **MUST** be a **Microsoft Word** file [Compulsory for text-based submission to enable Turnitin Plagiarism Check] or opt for PDF and IMAGE file [for non-text/ hand written submission], in A4 size format. **Note: Please keep the file size NOT exceeding 30MB per file.**

6. Please check your Index Number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You MUST name your answer script file using the following file name for submission:
   [UECS3253]_FA_[Programme Abbreviation]_[Your Index Number] or
   [UECS3463]_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme IB, and your Index Number is A99999CBIBF, then your answer script should be named as:
UECS3253_ FA_ IB_A99999CBIBF.doc

Answer Script File Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.
   (a).Attach your answer script at WBLE
   (b).Send your answer script to UECS3253.LKCFES@utar.edu.my/ UECS3463.LKCFES@utar.edu.my
   Note: For the subject / title of your email, use the file name of your answer script, i.e.,
   UECS3253_FA_[Programme Abbreviation]_[Your Index Number]

8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s).

Contents of Answer Script

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
   - Your Programme (Abbreviation)
   - Your Index Number
   - Your Name
   - Your Student ID

10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.

11. Each question should be answered starting on a new page. It is recommended that the answer to each question is limited to **[no. of] pages or [no. of] words**, whichever is lower. [Format / word Limit / Page Limit]

12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size **[e.g. 12]**, with proper spacing and alignment, except for the drawings and equations/calculations.

13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the

13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.

14. Please include a page number on each and every page of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UECS3253/ UECS3463 WIRELESS APPLICATION DEVELOPMENT

Answer **ALL THREE (3)** questions.

Q1. (a) What is React Native? (6 marks)

(b) React-Native is very similar to ReactJS in a way, but there are differences need to know before you start your first native app. Explain **FOUR (4)** differences between mobile development with React Native and web application development with ReactJS. (12 marks)

(c) Mobile apps can be classified into three categories which are native apps, mobile web apps and hybrid apps. Explain **THREE (3)** differences between native apps and mobile web apps. You may include the drawing of the illustration of the mobile apps. (15 marks)

[Total : 33 marks]

Q2. Table 2.1 shows the sample data of a table named **ebookpages** in an SQLite database named **favoritebookDb** that is used in the mobile application providing a listing of all the favorite books on the E-book pages.

id	bookid	book_name	author_name
1	P0023	Phython	Iris Smith
2	A1298	How to code C++	Jane Doe
3	W6522	Mobile Application	Zachery

Table 2.1: Sample data of the ebook pages

(a) Data of E-book Pages are downloaded by the mobile app to be inserted into the above mentioned **ebookpages** table upon first launch of the app. The data is in JSON format and is available in a variable named **favoritebookPages**. Figure 2.2 shows the sample data in JSON format.

```
[{
"bookid ": " P0023",
"book_name ": " Phython ",
"author_name ": " Iris Smith "
},
{
"bookid ": " A1298 ",
"book_name ": "How to code C++",
"author_name": "Jane Doe"
},
{
"bookid ": " W6522",
"book_name ": "Mobile Application ",
"author_name": "Zachery"
},]
```

Q2(a). (Continued)

Construct code segment to insert new rows to the ebookpages table using the JSON data in the E-book pages variable. (10 marks)

(b) Updates to the data about the E-book pages are downloaded by the app to be updated into the ebookpages table. The data is in JSON format as shown in the sample in Figure 2.2 and is available in a variable named pagesUpdate. Construct code segment to insert or update the rows in ebookpages using the JSON data in the pagesUpdate variable. If the bookid is already exist, perform an update. Otherwise, insert a new row. (16 marks)

(c) A screen of the mobile app displays a FlatList listing the names of all Facebook Pages. Construct code segment to query the ebookpages table and prepare the data as a JavaScript object that is suitable to populate the FlatList with the names of the Facebook Pages. (7 marks)

[Total : 33 marks]

Q3. KlangvalleyTIS is a mobile app that provides real-time traffic information in Klang Valley. This app is required to make a TCP socket connection to a server at **klangvalleytis.my: 1758** to send and receive data between the app and the server.

(a) Users of the app are required to turn on location services as the app needs to send the current location information to the server. For the purpose of sending location data, the app sends it in JSON format as shown in the example in Figure 3.1.

```
{
        "user_id ": "2977" ,
        "lat": 3.0390591 ,
        "lon": 101.7936234
}
```

Figure 3.1: Sample location data

The server responds to the app in JSON format as shown in the example in Figure 3.2 whether the location update is successful.

Q3(a). (Continued)

```
{
        "user_id ": "2977" ,
        "lat": 3.0390591 ,
        "lon": 101.7936234 ,
        "update ": true
}
```

Figure 3.2: Sample server response

Construct code segment to open a TCP connection to the **WebSocket** URL at **klangvalleytis.my:1758/update** using the **WebSocket** API and send the user sign in data as shown in Figure 3.1 to the server. If the location in is successful, invoke a method named **updateLocationOnMap**(). Otherwise, ignore the error.

If a connection error occurs, display an alert with an appropriate error message. Assume that the JSON data in Figure 3.1 is already set in a state named user loc.

(22 marks)

(b) The server program sends traffic information to the mobile app every 30 seconds. The traffic information is sent in JSON format as shown in the example in Figure 3.3. Upon receiving the traffic information, the method **updateTrafficOnMap()** will be called to update the traffic information on the map display.

```
[
    {
        "route_id ": 568,
        " traffic_class ": "B"
    },
    {
        "route_id ": 569,
        " traffic_class ": "A"
    }
]
```

Figure 3.3: Sample JSON response

Construct code segment to open a TCP connection to the **WebSocket** URL at **klangvalleytis.my:1758/status** to receive the traffic information and update the map.

(12 marks)

[Total : 34 marks]